

SAT UZBEKISTAN

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Module-1

1

Mark for Review



Which expression is equivalent to $(2x^2 + x - 9) + (x^2 + 6x + 1)$?

(A) $2x^2 + 7x + 10$

(B) $2x^2 + 6x - 8$

(C) $3x^2 + 7x - 10$

(D) $3x^2 + 7x - 8$

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Mark for Review



John paid a total of \$165 for a microscope by making a down payment of \$37 plus p monthly payments of \$16 each. Which of the following equations represents this situation?

(A) $16p - 37 = 165$

(B) $37p - 16 = 165$

(C) $16p + 37 = 165$

(D) $37p + 16 = 165$

3

Mark for Review



$$7m = 2(n + p)$$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives m in terms of n and p ?

(A) $m = \frac{2(n+p)}{7}$

(B) $m = 2(n + p)$

(C) $m = 2(n + p) - 7$

(D) $m = 2 - n - p - 7$

The function g is defined by $g(x) = \sqrt{8x + 1}$. What is the value of $g(3)$?

- (A) $\frac{5}{8}$
- (B) $\frac{25}{8}$
- (C) 5
- (D) 25

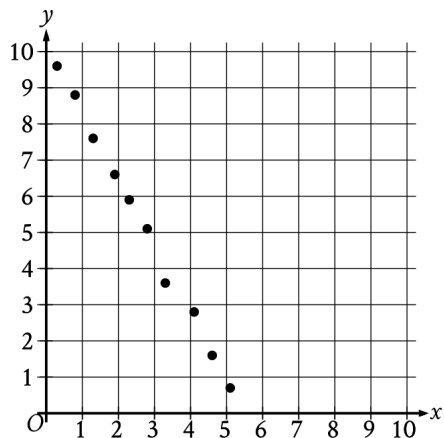
The table gives the distribution of votes for a new school mascot and grade level for 80 students.

Mascot	Grade level			
	Sixth	Seventh	Eighth	Total
Badger	4	9	9	22
Lion	9	2	9	20
Longhorn	4	6	4	14
Tiger	6	9	9	24
Total	23	26	31	80

If one of these students is selected at random, what is the probability of selecting a student whose vote for new mascot was for a lion?

- (A) $\frac{1}{9}$
- (B) $\frac{1}{5}$
- (C) $\frac{1}{4}$
- (D) $\frac{2}{3}$

A student council group is selling school posters for a fundraiser. They use the function $p(x) = 5x - 220$ to determine their profit $p(x)$, in dollars, for selling x school posters. In order to earn a profit of \$900, how many school posters must they sell?



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

(A) $y = -1.9x - 10.1$

(B) $y = -1.9x + 10.1$

(C) $y = 1.9x - 10.1$

(D) $y = 1.9x + 10.1$

$$3x + 6 = 4y$$

$$3x + 4 = 2y$$

The solution to the given system of equations is (x, y) . What is the value of y ?

Data value	Frequency
6	3
7	3
8	8
9	8
10	9
11	11
12	9
13	0
14	6

The frequency table summarizes the 57 data values in a data set. What is the maximum data value in the data set?

Circle K has a radius of 4 millimeters (mm). Circle L has an area of $100\pi\text{ mm}^2$. What is the total area, in mm^2 , of circles K and L ?

- (A) 14π
- (B) 28π
- (C) 56π
- (D) 116π

If $9(4 - 3x) + 2 = 8(4 - 3x) + 18$, what is the value of $4 - 3x$?

- (A) -16
- (B) -4
- (C) 4
- (D) 16



Triangle FGH is similar to triangle JKL , where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F) = \frac{308}{317}$, what is the value of $\sin(J)$?

(A) $\frac{75}{317}$

(B) $\frac{308}{317}$

(C) $\frac{317}{308}$

(D) $\frac{317}{75}$



A wire with a length of 106 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 6 more than 4 times the value of y . What is the value of x ?

(A) 25

(B) 28

(C) 56

(D) 86



A certain township consists of a 5-hectare industrial park and a 24-hectare neighborhood. The total number of trees in the township is 4,529. The equation $5x + 24y = 4,529$ represents this situation. Which of the following is the best interpretation of x in this context?

(A) The average number of trees per hectare in the industrial park

(B) The average number of trees per hectare in the neighborhood

(C) The total number of trees in the industrial park

(D) The total number of trees in the neighborhood

15

Mark for Review



Which expression is equivalent to $a^{\frac{11}{12}}$, where $a > 0$?

(A) $\sqrt[12]{a^{132}}$

(B) $\sqrt[144]{a^{132}}$

(C) $\sqrt[121]{a^{132}}$

(D) $\sqrt[11]{a^{132}}$

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Mark for Review

The function f is defined by $f(x) = (x - 6)(x - 2)(x + 6)$. In the xy -plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

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Mark for Review

$$\begin{aligned}y &= 4x + 1 \\4y &= 15x - 8\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x - y$?

18

Mark for Review



$$f(t) = 8,000(0.65)^t$$

The given function f models the number of coupons a company sent to their customers at the end of each year, where t represents the number of years since the end of 1998, and $0 \leq t \leq 5$. If $y = f(t)$ is graphed in the ty -plane, which of the following is the best interpretation of the y -intercept of the graph in this context?

(A) The minimum estimated number of coupons the company sent to their customers during the 5 years was 1,428.

(B) The minimum estimated number of coupons the company sent to their customers during the 5 years was 8,000.

(C) The estimated number of coupons the company sent to their customers at the end of 1998 was 1,428.

(D) The estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

A landscaper uses a hose that puts $88x$ ounces of water in a bucket in $5y$ minutes. Which expression represents the number of ounces of water the hose puts in the bucket in $9y$ minutes at this rate?

(A) $\frac{9x}{440}$

(B) $\frac{440x}{9}$

(C) $\frac{5x}{792}$

(D) $\frac{792x}{5}$

$$\sqrt{(x-2)^2} = \sqrt{3x+34}$$

What is the smallest solution to the given equation?

Triangle XYZ is similar to triangle RST such that X , Y , and Z correspond to R , S , and T , respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

(A) 2°

(B) 10°

(C) 20°

(D) 40°

$$f(x) = 9(4)^x$$

The function f is defined by the given equation. If $g(x) = f(x+2)$, which of the following equations defines the function g ?

(A) $g(x) = 18(4)^x$

(B) $g(x) = 144(4)^x$

(C) $g(x) = 18(8)^x$

(D) $g(x) = 81(16)^x$

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Module-2

1

Mark for Review

73, 74, 75, 77, 79, 82, 84, 85, 91

What is the median of the data shown?

2

Mark for Review



Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

(A) 37°

(B) 53°

(C) 127°

(D) 143°

3

Mark for Review



x	y
1	11
2	19
3	a

The table shows three values of x and their corresponding values of y for the equation $y = 4(2)^x + 3$. In the table, a is a constant. What is the value of a ?

(A) 67

(B) 35

(C) 32

(D) 27



$$66x = 66x$$

How many solutions does the given equation have?

(A) Exactly one

(B) Exactly two

(C) Infinitely many

(D) Zero



A model predicts that the population of Bergen was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

(A) $p = 0.96(15,000)^x$

(B) $p = 1.04(15,000)^x$

(C) $p = 15,000(0.96)^x$

(D) $p = 15,000(1.04)^x$



The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the xy -plane has an x -intercept at $(a, 0)$ and a y -intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$?

(A) 21

(B) 28

(C) 32

(D) 35

7

Mark for Review



Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x > 7$?

(A) $\frac{x-7}{5}$

(B) $\frac{8x-3}{2}$

(C) $\frac{8x^2-3x-14}{2x-14}$

(D) $\frac{8x^2-3x-77}{2x-14}$

8

Mark for Review



Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A ?

(A) 43

(B) 86

(C) 129

(D) 1,849

9

Mark for Review

A circle has center O , and points R and S lie on the circle. In triangle ORS , the measure of $\angle ROS$ is 88° . What is the measure of $\angle RSO$, in degrees? (Disregard the degree symbol when entering your answer.)

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Mark for Review



A business owner plans to purchase the same model of chair for each of the 81 employees. The total budget to spend on these chairs is \$14,000, which includes a 7% sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

(A) \$148.15

(B) \$161.53

(C) \$172.84

(D) \$184.94

11

[Mark for Review](#)

A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

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[Mark for Review](#)


The positive number a is 230% of the number b , and a is 60% of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

(A) 138

(B) 217

(C) 283

(D) 383

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[Mark for Review](#)


For $x > 0$, the function f is defined as follows:

$f(x)$ equals 201% of x

Which of the following could describe this function?

(A) Decreasing exponential

(B) Decreasing linear

(C) Increasing exponential

(D) Increasing linear

14

[Mark for Review](#)

x	y
k	13
$k + 7$	-15

The table gives the coordinates of two points on a line in the xy -plane. The y -intercept of the line is $(k - 5, b)$, where k and b are constants. What is the value of b ?

Keenan made 32 cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

- (A) The number of large jars Keenan filled
- (B) The number of small jars Keenan filled
- (C) The total number of cups of vegetable broth in the large jars
- (D) The total number of cups of vegetable broth in the small jars

$$3x = 36y - 45$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

- (A) $x = 4y$
- (B) $\frac{1}{3}x = 4y$
- (C) $x = 12y - 15$
- (D) $\frac{1}{3}x = 12y - 15$

$$x - 29 = (x - a)(x - 29)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 30$?

- I. a
- II. $a + 1$
- III. 29

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II, and III

18

Mark for Review



One gallon of stain will cover 170 square feet of a surface. A yard has a total fence area of w square feet. Which equation represents the total amount of stain S , in gallons, needed to stain the fence in this yard twice?

(A) $S = \frac{w}{170}$

(B) $S = 170w$

(C) $S = 340w$

(D) $S = \frac{w}{85}$

19

Mark for Review



In the xy -plane, the graph of the equation $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c ?

(A) $-\frac{481}{4}$

(B) -100

(C) $-\frac{319}{4}$

(D) $-\frac{9}{2}$

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Mark for Review

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water and then reached the surface of the water 12 minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water?



The area of a rectangular region is increasing at a rate of 250 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

(A) 0.39

(B) 1.27

(C) 13.67

(D) 23.24



$$5x + 7y = 1$$

$$ax + by = 1$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

(A) $10x + 7y = 1$
 $ax - 2by = 1$

(B) $10x + 7y = 1$
 $ax + 2by = 1$

(C) $10x + 7y = 1$
 $2ax + by = 1$

(D) $5x - 7y = 1$
 $ax + by = 1$